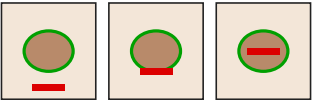


(a) Controlled sweep

synthetic artifact, controlled overlap r



$r = 0$ $r = 0.5$ $r = 1$

artifact moves into the ROI

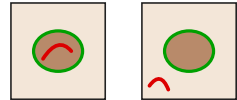
ROI (lesion, nodule, tumour, lung)

artifact (ruler, hair, caliper, debris)

removed by masking

(b) Real traps

different images carry the artifact



Trap A
 $r \geq 0.5$

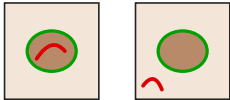
Trap B
 $r < 0.1$



shared artifact-free group

(c) Transplant

one real artifact, same image, two places



inside
 $r \geq 0.95$

outside
 $r = 0$



control: neutral tissue through the same mask

(d) Arms and environments



ERM
(whole image)



ROI masking
(outside removed)

artifact rate in $Y = 1$ / $Y = 0$

train	0.9 / 0.1
correlated test	0.9 / 0.1
reversed test	0.1 / 0.9
clean test	independent

location crossover = $[\text{mask} - \text{ERM}]_{\text{outside}} - [\text{mask} - \text{ERM}]_{\text{inside}}$ (reversed-test AUROC)

positive when masking removes a shortcut outside the ROI but not one inside it